United States Department of the Interior

BUREAU OF RECLAMATION
P.O. Box 25007
Denver, CO 80225-0007

IN REPLY REFER TO:
86-68540
2.2.4.21

VIA ELECTRONIC MAIL ONLY

MEMORANDUM

To: Sara Putnam, P.E., Civil Engineer
Plant Structures (86-68120)
Technical Service Center, Denver, CO

From: Daryl Little, Ph.D., Supervisory Civil Engineer
Materials and Corrosion Laboratory Group (MCL) (86-68540)
Technical Service Center, Denver, CO

DARYL LITTLE
Digitally signed by DARYL LITTLE
Date: 2026.10.08 20:12:34 -06'00'

Subject: Soil Corrosivity Testing – San Juan Lateral – Turnout Buildings, Corrosivity Report, Upper Colorado Basin Region, Navajo-Gallup Water Supply Project, New Mexico. Referral Number: 8540-2026-19

INTRODUCTION

Corrosivity testing was performed on 13 soil samples collected from Test Pits near the turnout building locations. Samples were tested at Weld Laboratories, Inc in Greeley, Colorado for resistivity, pH, and soluble chloride and sulfate concentrations.

CONCLUSIONS

- Resistivity values fall in the “very severe” range for 6 samples and “severe” range for 6 samples.
- pH values fall in the “neutral” range for all 13 samples.
- Chloride concentrations fall in the “low” range for 10 samples.
- Sulfate concentrations fall in the “low” range for 9 samples, but 3 samples fall into the severe range.
- 12 of the 13 samples indicate a highly corrosive environment; it is recommended that buried ferrous materials be specified with a protective coating and cathodic protection.
- Soil chemistry can vary with location; additional testing may be needed to make conclusions about the corrosivity of the soil in other project locations. TSC-MCL staff have expertise in corrosion mitigation.
- These results indicate need for special considerations on this project pertaining to corrosion of metallic structures and concrete structures. TSC Concrete and Structural Laboratory (CSL) staff have expertise in concrete mix design. This group can provide additional interpretation and support, if needed.

CORROSIVITY TESTING AND RESULTS

The purpose of corrosivity testing is to quantify properties of soil that indicate a high probability for adverse reaction with buried structures, such as corrosion of metals or sulfate attack in concrete. Such properties include the concentrations of water-soluble minerals that may be dissolved and transported to the structure during periods of high precipitation. Reclamation recommends corrosivity testing for any structure, metallic or concrete, which comes in contact with soil. Corrosivity testing consists of measuring soil resistivity, pH, and the concentrations of water-soluble anions, with primary analytes being chloride (Cl^-) and sulfate (SO_4^{-2}).

Samples were tested at third-party laboratory, Weld Laboratories, Inc. The standards used by this laboratory for corrosivity testing are:

- Resistivity – AASHTO T288-12
- pH – AASHTO T289-91
- Chlorides – AASHTO T291-94
- Sulfates – AASHTO T290-95

Resistivity values fall in the "moderate" range for 1 sample, "severe" range for 6 samples, and "very severe" range for 6 samples. The measured values for pH are within the "neutral" range for all samples. Chloride concentrations fall in the "low range" for 10 samples and the "moderate" range for 3 samples:

Chlorides Low	Chlorides Moderate
TPTB-25-(1-4, 7, 9-12, and 23)	TPTB-25-(5, 6, and 8)

Sulfate concentrations fall in the "low" range for 9 samples, the "moderate" range for 1 samples, and the "severe" range for 3 sample:

Sulfates Low	Sulfates Moderate	Sulfates Severe
TPTB-25-(3, 4, 7-12, and 23)	TPTB-25-6	TPTB-25-(1, 2, and 5)

Test results are included as Appendix A.

The following table specifies the accompanying test values with their respective range.

Table I. Test Ranges and Values

	Range	Values
Resistivity (Ω-cm)	Low	>10,000
	Moderate	5,000-10,000
	Severe	2,000-5,000
	Very Severe	<2,000
pH	Acidic	<6
	Neutral	6-9
	Basic	>9
Chlorides (mg/L)*	Low	<100
	Moderate	100-500
	Severe	>500
Sulfates (mg/L)*	Low	<1,000
	Moderate	1,000-2,000
	Severe	2,000-20,000
	Very Severe	>20,000

* 1 mg/L ≈ 1 ppm = 0.0001 %

cc: FCCO-725 (C. Beyer), 86-68320 (T. Lewis), 86-68540 (D. Little), 86-68550 (T. Chatfield)

APPENDIX A

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-1
Laboratory No.: E25219-5A

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	7.5	0
Conductivity (mmhos/cm)	1.610	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	6.21	
Minimum Lab Resistivity (ohm-cm)	370	10
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	3.7	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	98	3.5
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	0.5	2
Chloride (mg/kg DMB) AASHTO T 291-94	63	1
Sulfate (mg/kg DMB) AASHTO T 290-95	4003	3
Sulfate-S (mg/kg DMB)	1334	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

9/13/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-2
Laboratory No.: E25219-5B

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	7.7	0
Conductivity (mmhos/cm)	1.056	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	9.47	
Minimum Lab Resistivity (ohm-cm)	420	10
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	4.2	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	87	3.5
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	1	2
Chloride (mg/kg DMB) AASHTO T 291-94	59	1
Sulfate (mg/kg DMB) AASHTO T 290-95	3632	3
Sulfate-S (mg/kg DMB)	1211	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

9/13/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-3
Laboratory No.: E25219-5C

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	8.0	0
Conductivity (mmhos/cm)	0.555	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	18.02	
Minimum Lab Resistivity (ohm-cm)	740	10
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	7.4	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	94	3.5
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	1	2
Chloride (mg/kg DMB) AASHTO T 291-94	63	1
Sulfate (mg/kg DMB) AASHTO T 290-95	244	3
Sulfate-S (mg/kg DMB)	81	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

9/13/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-4
Laboratory No.: E25219-5D

	Results[1,3]	10-Point System[2]
pH (SI)	8.3	0
AASHTO T 289-91 (ASTM G51 available for some soil)		
Conductivity (mmhos/cm)	0.334	NA
Resistivity (ohm-m)	29.94	
USDA Handbook 60, temperature corrected conductivity probe		
Minimum Lab Resistivity (ohm-cm)	1090	10
Minimum Lab Resistivity (ohm-m)	10.9	
via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]		
Redox (mV vs. Ag/AgCl)	79	3.5
ASTM G200 (ASTM D1498 if soil is low in moisture)		
Free Sulfide (mg/kg DMB)	0.5	2
EPA 9030B+9034, prescreened with lead acetate paper		
Chloride (mg/kg DMB)	34	0
AASHTO T 291-94		
Sulfate (mg/kg DMB)	166	1
AASHTO T 290-95		
Sulfate-S (mg/kg DMB)	55	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The CI- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

9/13/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-5
Laboratory No.: E25219-5E

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	8.1	0
Conductivity (mmhos/cm)	2.160	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	4.63	
Minimum Lab Resistivity (ohm-cm)	140	10
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	1.4	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	62	3.5
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	ND	0
Chloride (mg/kg DMB) AASHTO T 291-94	113	3
Sulfate (mg/kg DMB) AASHTO T 290-95	2061	3
Sulfate-S (mg/kg DMB)	687	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The CI- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

9/13/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-6
Laboratory No.: E25219-5F

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	8.0	0
Conductivity (mmhos/cm)	2.020	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	4.95	
Minimum Lab Resistivity (ohm-cm)	190	10
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	1.9	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	59	3.5
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	ND	0
Chloride (mg/kg DMB) AASHTO T 291-94	104	3
Sulfate (mg/kg DMB) AASHTO T 290-95	1512	3
Sulfate-S (mg/kg DMB)	504	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

John Willis
Project Manager

9/13/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-7
Laboratory No.: E25219-5G

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	8.9	3
Conductivity (mmhos/cm)	0.291	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	34.36	
Minimum Lab Resistivity (ohm-cm)	2390	2
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	23.9	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	142	0
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	ND	0
Chloride (mg/kg DMB) AASHTO T 291-94	15	0
Sulfate (mg/kg DMB) AASHTO T 290-95	55	0
Sulfate-S (mg/kg DMB)	18	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

9/17/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-8
Laboratory No.: E25219-5H

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	8.7	3
Conductivity (mmhos/cm)	0.412	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	24.27	
Minimum Lab Resistivity (ohm-cm)	121	10
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	1.21	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	123	0
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	ND	0
Chloride (mg/kg DMB) AASHTO T 291-94	106	3
Sulfate (mg/kg DMB) AASHTO T 290-95	68	0
Sulfate-S (mg/kg DMB)	23	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-9
Laboratory No.: E25219-5 I

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	8.7	3
Conductivity (mmhos/cm)	0.267	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	37.45	
Minimum Lab Resistivity (ohm-cm)	1890	5
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	18.9	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	64	3.5
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	ND	0
Chloride (mg/kg DMB) AASHTO T 291-94	30	0
Sulfate (mg/kg DMB) AASHTO T 290-95	123	1
Sulfate-S (mg/kg DMB)	41	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

9/13/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-10
Laboratory No.: E25219-5J

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	8.3	0
Conductivity (mmhos/cm)	0.483	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	20.70	
Minimum Lab Resistivity (ohm-cm)	570	10
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	5.7	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	64	3.5
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	ND	0
Chloride (mg/kg DMB) AASHTO T 291-94	53	1
Sulfate (mg/kg DMB) AASHTO T 290-95	117	1
Sulfate-S (mg/kg DMB)	39	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

9/13/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-11

Laboratory No.: E25219-5K	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	8.8	3
Conductivity (mmhos/cm)	0.109	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	91.74	
Minimum Lab Resistivity (ohm-cm)	3360	0
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	33.6	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	88	3.5
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	ND	0
Chloride (mg/kg DMB) AASHTO T 291-94	17	0
Sulfate (mg/kg DMB) AASHTO T 290-95	22	0
Sulfate-S (mg/kg DMB)	7	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

Date

Sampling procedures can affect the value of analytical results - customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

September 13, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPTB-25-12
Laboratory No.: E25219-5L

	Results[1,3]	10-Point System[2]
pH (SI) AASHTO T 289-91 (ASTM G51 available for some soil)	8.4	0
Conductivity (mmhos/cm)	0.110	NA
Resistivity (ohm-m) USDA Handbook 60, temperature corrected conductivity probe	90.91	
Minimum Lab Resistivity (ohm-cm)	3810	0
Minimum Lab Resistivity (ohm-m) via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]	38.1	
Redox (mV vs. Ag/AgCl) ASTM G200 (ASTM D1498 if soil is low in moisture)	106	0
Free Sulfide (mg/kg DMB) EPA 9030B+9034, prescreened with lead acetate paper	ND	0
Chloride (mg/kg DMB) AASHTO T 291-94	16	0
Sulfate (mg/kg DMB) AASHTO T 290-95	20	0
Sulfate-S (mg/kg DMB)	7	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.

WELD LABORATORIES, INC.

1527 First Avenue • Greeley, Colorado 80631
Phone: (970) 353-8118 • Fax: (970) 353-1671
www.weldlabs.com

August 27, 2025

Bureau of Reclamation
Attn: M Oligney 86-68550
PO Box 25007 MS 86-68540
Denver, CO 80225

Project No.: Navajo Gallup Water Supply Project
Sample ID: TPR2-15-23

Laboratory No.: E25219-6	Results[1,3]	10-Point System[2]
pH (SI)	8.7	3
AASHTO T 289-91 (ASTM G51 available for some soil)		
Conductivity (mmhos/cm)	0.154	NA
Resistivity (ohm-m)	64.94	
USDA Handbook 60, temperature corrected conductivity probe		
Minimum Lab Resistivity (ohm-cm)	5430	0
Minimum Lab Resistivity (ohm-m)	54.3	
via Miller Box, Tinker & Razor SR-2 (AASHTO T 288-12)[4]		
Redox (mV vs. Ag/AgCl)	71	3.5
ASTM G200 (ASTM D1498 if soil is low in moisture)		
Free Sulfide (mg/kg DMB)	0.1	2
EPA 9030B+9034, prescreened with lead acetate paper		
Chloride (mg/kg DMB)	19	0
AASHTO T 291-94		
Sulfate (mg/kg DMB)	44	0
AASHTO T 290-95		
Sulfate-S (mg/kg DMB)	15	

1. NA = Not Analyzed; ND = Not Detected. DMB = Dry Matter Basis. Measurements taken at 25°C.
2. 10-point Corrosion system based on: Appendix A of ANSI/AWWA C105/A21.5 Standard "Polyethylene Encasement for Ductile Iron Pipe Systems." The Cl- points adapted from the DIPRA design decision model. Sulfate is penalized at half the rate of chloride: A. A. Sagüés et. al. (https://rosap.ntl.bts.gov/view/dot/17493)
3. pH, Conductivity, and Redox are generally read on a 1:1 soil:water mixture if the soil is dry.
4. ASTM G57 4-Electrode Method used unless 2-electrode method is requested.

Project Manager

8/27/2025
Date

Sampling procedures can affect the value of analytical results – customers are advised to use appropriate sampling protocol to ensure samples are truly representative of the bulk sample.